

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated
for lumbosacral transitional anatomy**

[Author names removed for double-anonymized review]

[Affiliations removed for double-anonymized review]

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Purpose: Lumbar levels cannot be numbered reliably when a lumbosacral transitional vertebra (LSTV) is present. Levels are conventionally established by counting downward from C2, the reference standard for numeration¹. An abdominal field of view lacks that landmark, leaving a specific ambiguity: four rib-free lumbar vertebrae may mean an L1 bearing a lumbar rib or an L5 assimilated to the sacrum, and six may mean a true sixth lumbar vertebra or a T12 whose ribs are aplastic². CT-SpinoPelvic1K is built to test whether local morphology can resolve them without counting from the C2 anchor. It provides 802 CT records anchoring each lumbar level on the lowest rib-bearing vertebra above and the first sacral segment below, with radiologist-sourced spine *and* pelvic annotation on one frame, per-level ribs and femora. Classes for a sixth lumbar vertebra, a thirteenth thoracic vertebra, a separate first sacral segment, and lumbar ribs make the set expressive enough for anatomical anomalies.

Acquisition and Validation Methods: Records pair TCIA CT COLONOGRAPHY imaging with CTSpine1K³ vertebral and CTPelvic1K⁴ pelvic labels on the acquisition with highest bone coverage, under a VerSe-native scheme. Validation: geometric invariants (802/802 pass); rib-vertebra incidence across 5,749 evaluable ribs (2 offsets, 0.035%); and agreement of spinopelvic measures with published values.

Data Format and Usage Notes: Gzipped NIfTI image/label pairs sharing one affine, canonicalized to PIR, with frozen patient-grouped LSTV-stratified five-fold splits and a reference loader; archived at 10.5281/zenodo.22139642.

Potential Applications: Testing whether a network can classify a vertebra from local features alone; updating textbook morphometry drawn from small cadaveric series; spinopelvic assessment at the lumbosacral interface; variant studies; opportunistic screening. *Limitations:* thoracic ground truth is field-of-view limited; postural angles supine; no held-out test set; the rib layer is a triaged-review pseudolabel; Castellvi grades read independently by two radiology resident physicians and resolved by consensus.

6 I. INTRODUCTION

7 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and
8 a vertebra is identified by where it falls in that sequence. Transitional variants sit at the two
9 ends of the lumbar column — the thoracolumbar transitional vertebra (TLTV) above, the
10 lumbosacral transitional vertebra (LSTV) below — and they disturb identification in two
11 ways at once. They change what the border vertebra and its ribs *look like*: a thoracolumbar
12 border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral
13 one may be partly or completely assimilated to the sacrum, with an anteriorly wedged body,
14 a reduced disc beneath it and hypoplastic facet joints, or conversely separated from it with
15 a squared body, lumbar-type facets and a full-height disc². They also change *how many*
16 vertebrae each region contains — eleven or thirteen thoracic, four or six lumbar.

17 Either change alone would be tractable. Together they make it difficult to state defini-
18 tively what any vertebra at these borders *is*. Four rib-free vertebrae above the sacrum may
19 mean an L1 bearing a lumbar rib or an L5 assimilated to it; six may mean a true sixth
20 lumbar vertebra or a T12 whose ribs are aplastic. Each pair yields the same count, and the
21 morphology that would separate the two readings is not decisive to a human reader in every
22 case. Whether it nonetheless differs enough for a learned classifier to separate them, or at
23 least to assign each reading a probability, is the question this release is built to support.
24 Absent that, what remains is the sequence — conventionally established by counting down-
25 ward from C2, the reference standard for numeration¹, and unavailable in any spine-limited
26 acquisition. LSTV is common, reported between 4% and 30% depending on definition, and
27 wrong-level surgery is its most serious consequence.

28 Existing public collections cannot address this, and size is not the reason (Table I). Spine
29 datasets omit the pelvis; pelvic datasets do not number the vertebrae; and TotalSegmentator,
30 the whole-body scheme, has no class for a sixth lumbar vertebra or for a rib on a lumbar
31 vertebra. A six-lumbar spine therefore cannot be recorded in any of them, however good
32 the segmentation. VerSe^{5,6} makes the point most clearly (Table I): it deliberately enriched
33 for anatomical variants and graded them by Castellvi, yet left the graded vertebrae fused to
34 the sacrum unsegmented.

35 CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and
36 gives a class to each structure an enumeration anomaly produces: a sixth lumbar vertebra,

37 and a rib borne by a lumbar vertebra. A scheme without those classes must record such
38 anatomy as something it is not.

39 *Where the material is.* Everything described here is v7 of a single archived dataset, de-
40 posited at [DOI redacted for review]. Citations should use [DOI redacted for review], the per-
41 manent identifier for the dataset across all of its versions, which resolves to whichever version
42 is current; the version identifier above pins the exact release measured in this manuscript.
43 Documentation, the label scheme, the reference loader and the per-record quality-control
44 tables are distributed with the archive and are described in Sec. [III](#).

45 **I.A. Vertebra labeling when the scan cannot settle the count**

46 *The limitation, stated plainly.* No scan in this corpus contains C2 or the whole cervical
47 spine, so the conventional count cannot be performed. That is a property of abdominal
48 imaging rather than of the annotation, and it means a variant at the thoracolumbar junction
49 cannot be *resolved by counting* here at all: a thirteenth thoracic vertebra, a lumbar rib, and
50 a *stump rib* — a hypoplastic twelfth rib, one of the cardinal indicators of a thoracolumbar
51 transitional vertebra — produce overlapping appearances.

52 *What the release does instead.* No vertebral label is asserted where it cannot be supported.
53 Every quantity is expressed against the two landmarks present in essentially every abdominal
54 scan — the sacrum below and the lowest rib-bearing vertebra above — so the description is
55 positional rather than nominal: a vertebra by how many rib-free vertebrae separate it from
56 the sacrum, a lowest rib by its length as a fraction of the rib above it, a transverse process
57 by its span and its distance from the ala. None of these changes according to whether a
58 reader would have said L5 or S1, so cases that disagree about the name remain comparable.
59 Where the junction is in view the count is determined and the conventional name follows,
60 and the two descriptions agree. The positional description is, by construction, blind to the
61 ambiguity: an L1 bearing a lumbar rib and an L5 assimilated to the sacrum both leave
62 four rib-free vertebrae above the sacrum, and a true sixth lumbar vertebra and a T12 with
63 aplastic ribs both leave six. The count alone cannot separate the readings.

64 *But the vertebrae themselves are known to differ.* Counting is not the only route to an
65 identity. Level-specific morphometry is long established: pedicle width and height change
66 systematically from the thoracic to the lumbar spine,⁷ vertebral body, endplate and canal

TABLE I Public CT collections at the lumbosacral junction: size, and what each published label scheme can represent. Teal: class present. Amber: Castellvi anatomy is graded, but only to exclude it — vertebrae partially fused to the sacrum are left unsegmented⁶, 21 of the 33 graded records here. Three comparators carry more scans and RibSeg v2¹² labels more ribs; the claim is the class set, not the size.

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	Lumbar rib	Femora	Grade
CTSpine1K ³	1,005									
CTPelvic1K ⁴	1,184									
VerSe ⁵	374									
RibSeg v2 ¹²	660									
TotalSegmentator ¹⁶	1,204									
CTSpinoPelvic1K	802									

67 dimensions differ by level,^{8–10} and the transverse process and the iliolumbar ligament mark
 68 the last lumbar vertebra independently of any count.^{2,11} If those differences hold at the
 69 boundary itself, the phenotype is decidable from the vertebra in front of the reader without
 70 any global count. On this corpus it does hold: separating T12 from L1 on shape alone, with
 71 each patient’s own size divided out and cross-validation grouped by case, gives an area under
 72 the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is deliberately removed,
 73 because a classifier given raw millimeters separates thoracic from lumbar immediately and
 74 has learned nothing that helps at the junction, where the two are nearly the same size.

75 That is reported as a property of the released measurements rather than as a method:
 76 the information needed to settle these variants is present locally.

77 II. ACQUISITION AND VALIDATION METHODS

78 II.A. Sources, and why a crosswalk was necessary

79 Both CTSpine1K³ and CTPelvic1K⁴ draw part of their imaging from the TCIA CT
 80 colonography collection^{13,14}, and the vertebral and pelvic annotations in both were pro-
 81 duced under board-certified radiologist supervision. Neither released the mapping from an
 82 annotation to the specific CT series it was drawn on.

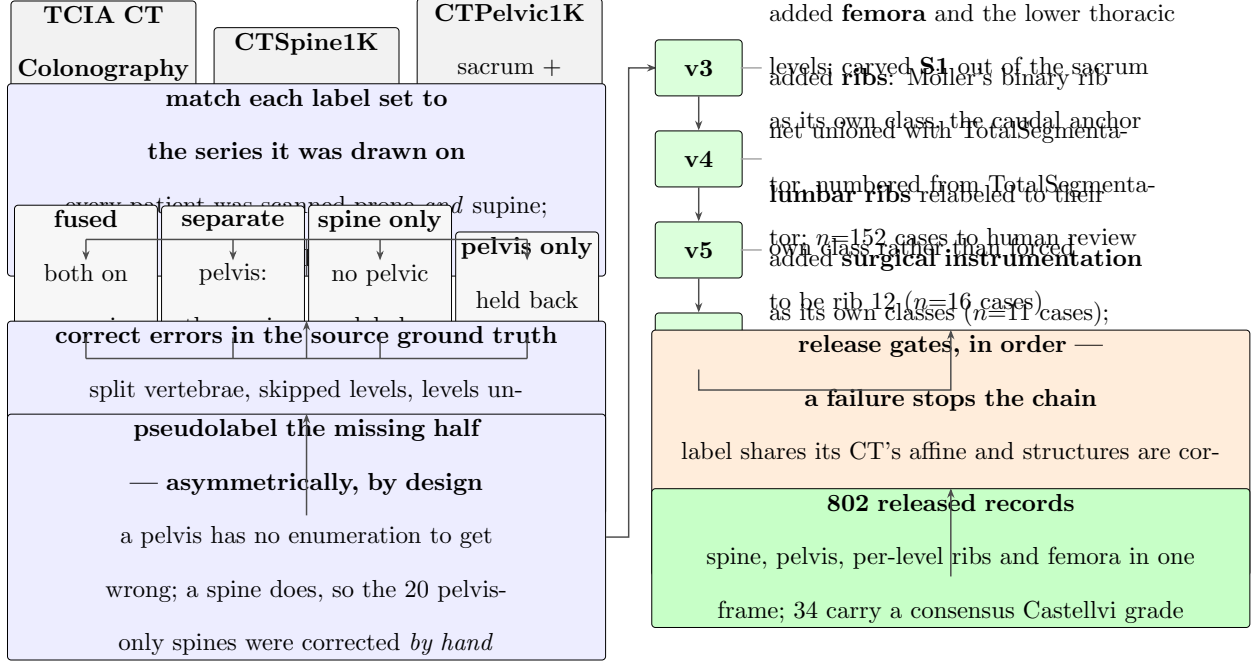


FIG. 1 How the dataset was built. Three public collections are combined patient by patient; neither annotation source published the CT series its labels were drawn on, so each label set is matched to the series whose bone agrees with it, and in 351 patients the two sources had drawn on different scans of the same person. n counts scans, or corrections, at that step.

That mapping is not a bookkeeping detail. Every patient in CT COLONOGRAPHY was scanned *twice*, prone and supine, often with more than one kernel, so one patient identifier resolves to several volumes differing in posture, position, field of view and noise — and an annotation carries a patient identifier and nothing finer.

Placing it on the wrong volume of the same patient is not merely a displacement. The spine is mildly deformable, and turning a patient from supine to prone alters lumbar lordosis and the alignment of each segment against the next, so a vertebra moves by a different amount, and in a different direction, from the one above it. Outlines drawn on one series are the wrong *shape* for the other, and no rigid realignment recovers them. In isolation such a mask looks like competent work; it reveals itself only when overlaid on the image.

The crosswalk (Fig. 1) therefore resolves each annotation first to a patient and then, within that patient's volumes only, to the series whose bone *agrees* with it: candidates are thresholded at bone attenuation and scored by overlap with the annotation mask. Bone rather than image similarity, because bone is what the annotation describes — two acquisitions of one abdomen resemble each other everywhere except in where the skeleton sits.

98 II.B. Fused and split records

99 The two annotation sets were placed independently, so they do not always land on the
100 same series. Of 802 records, 342 carry spine and pelvic annotations on the same volume
101 (*fused*), 351 are *separate*, 89 are *spine-only* with no pelvic annotation for that patient, and
102 20 are *pelvic-only*. The split is the crosswalk’s main finding rather than bookkeeping: in
103 those separate records — nearly half of all patients — the two public collections annotated
104 *different volumes of the same person*, and neither recorded which. A pelvic annotation
105 placed on the other acquisition of a prone/supine pair cannot be combined with the spine
106 annotation without resampling across a change of posture. Those records are retained rather
107 than discarded, and held out of the postural analyses, because two acquisitions of one patient
108 differing in spinal alignment are a resource for a mobility study rather than a defect.

109 II.C. Densification, and why it is asymmetric

110 A corpus in which half the records lack a pelvis is not usable as a spinopelvic dataset,
111 so the missing structures were pseudolabeled — asymmetrically (Fig. 1), because the two
112 directions are not equivalent.

113 Pelvis onto spine-only records is the safe direction. The sacrum and innominate bones
114 are single objects with no enumeration: no count to get wrong, no naming convention to
115 choose, no analog of a transitional vertebra. Quality is reported as held-out performance on
116 the *pelvic-only* records, withheld from training for exactly this purpose.

117 The reverse direction carries the whole problem this dataset exists to address, since a
118 pseudolabeled spine must commit to a vertebral count and on a transitional vertebra will
119 commit to the commoner reading. Those 20 records were therefore corrected by hand rather
120 than densified automatically — tractable at twenty cases, which is the practical reason the
121 asymmetry runs the way it does.

122 II.D. Ribs

123 No public rib annotation covers this cohort, and the two available automatic sources fail
124 in complementary ways.

125 Möller’s binary rib network¹⁵ reports high accuracy, and its authors show that stump ribs

126 can be classified from a partial field of view; that result concerns the morphological features
127 computed on a rib once segmented. On this corpus the segmentation itself failed on ribs
128 that were *partially outside* the field of view: a rib crossing the boundary of an abdominal
129 acquisition was liable to be missed altogether rather than segmented to the edge. This is
130 not a limitation its authors report — their evaluation excluded subjects in which the last
131 rib was not visible, and their failure analysis attributes errors to costal-process fusion — so
132 it is recorded here as an observation from this cohort. Since this is a colonography cohort,
133 the lower ribs are routinely clipped, and those are precisely the ribs the count depends on.

134 TotalSegmentator¹⁶ does segment clipped ribs and supplies the per-rib numbering that a
135 binary network cannot. Its characteristic failure is at the other end of the bone: the most
136 medial portion of the rib — roughly the last tenth to sixth approaching the costovertebral
137 joint — is frequently absent. A rib truncated at its medial end never reaches the vertebra it
138 articulates with, so the rib–vertebra incidence check — the quality-control gate on the rib
139 class labels, which matches each rib to the vertebra its number implies — has nothing to
140 test.

141 The two were therefore unioned (Fig. 1): TotalSegmentator supplies numbering and the
142 medial-clipped cases, Möller supplies detection where its output is more complete, and the
143 union carries TotalSegmentator’s per-level identities.

144 The result is a pseudolabel whose review was triaged rather than exhaustive, which is
145 worth stating plainly because a layer described only as “human-corrected” invites the reader
146 to assume every bone was looked at. A label-based rule flagged every record in which one
147 rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it
148 articulates with — a single bone with two numbers being a numbering error by construction,
149 whatever the anatomy. The rule selected 152 records; 148 were reviewed and corrected, and
150 4 were not and are identified in the release. Reviewers were medical students working
151 through a student research consortium¹⁷, trained against a written labeling protocol, with
152 every correction recorded against the annotator who made it. Records passing the rule were
153 not individually inspected, so the layer’s guarantee is that it is free of the failure modes the
154 rule detects, and no stronger. The residual rib–vertebra offsets reported in Sec. II.G are the
155 independent check on what the rule missed.

II.E. Label scheme and counting anchors

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier above that range — S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs (34–45 left, 46–57 right), lumbar ribs (74–75) and surgical hardware (76–82). Identifiers 58–73 are unassigned: a soft-tissue block once reserved there was never populated, and the v7 descriptor still carries its names.

Two anchors make the count decidable, and Fig. 2 shows both: the lowest rib-bearing vertebra above, **S1** carved from the sacrum below. Each earns its place by what it supports rather than by being an endpoint — S1 supplies the sacral endplate landmark that sacral slope and pelvic incidence are measured from, and with both brackets present L5 and L6 are distinguishable where a spine-limited view alone would not be.

How S1 is cut. The sacrum’s outer boundary is the source annotation and is never altered; S1 is the cranial part of that same bone, separated by a plane. The plane’s normal is the sacrum’s own cranio-caudal axis, made orthogonal to a left–right axis that is measured rather than assumed — the normal of the reflection plane best mapping the sacrum onto itself, which removes the patient’s rotation within the scanner. That rotation is not negligible: median 4.5°, maximum 16.1°, and 508 of 801 records above 3°. The level of the cut preserves the sub-division volume of the preceding release, so the plane changes the shape and orientation of the S1/S2 boundary and not how much of the sacrum is called S1. Per-record geometry ships with the archive.

Neither anchor is novel: TotalSegmentator¹⁶ carries a separate S1 class and per-level ribs, and VerSe⁵ carries L6. Because this release uses the VerSe identifiers, its L6 *is* VerSe’s, interoperable rather than local to this work. L6 is retained not for novelty but because a scheme without it cannot record a six-lumbar spine at all, and must renumber the column or drop a level to fit.

A rib on a lumbar body receives its own class, and this is the one class here with no published counterpart. A lumbar rib and a stump rib are the same object under two counts, so assigning it a number would commit the label to one reading of an enumeration anomaly. A scheme that numbers every rib 1–12 has nowhere to put a thirteenth: the annotator must either call it rib 12 — asserting the vertebra beneath it is thoracic, the very question at

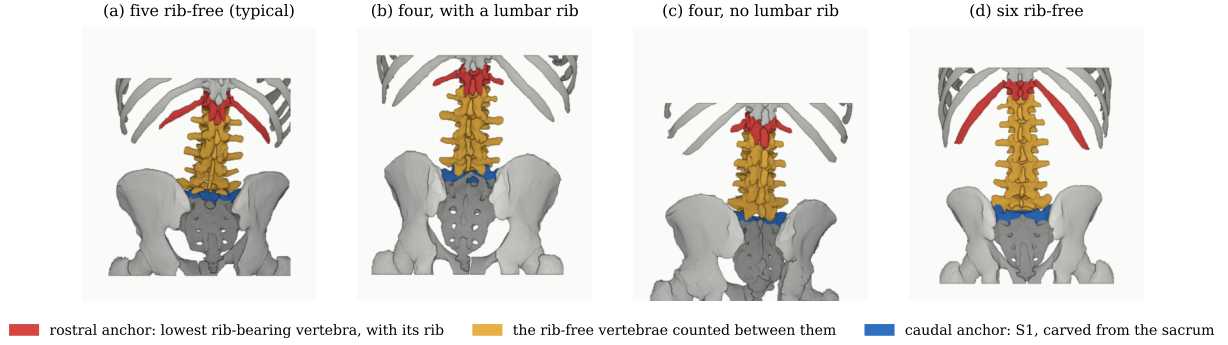


FIG. 2 The two anchors and the interval between them, from the released labels (posterior view). Red: the lowest rib-bearing vertebra with its rib. Blue: S1. Yellow: the rib-free vertebrae counted between. Colors follow role, not identifier. Panels (b) and (c) both count four and are indistinguishable by it. One millimeter scale, aligned at the sacral base.

187 issue — or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and
 188 VerSe’s T13 is a vertebra rather than a rib. 16 released records carry a lumbar rib and 18
 189 carry an L6.

190 Figure 2 shows why the interval is the primitive and the label is derived. Among the
 191 records with four rib-free vertebrae, some reach that count through a lumbar rib and some
 192 without one; the two are indistinguishable by the count alone and are distinguished here
 193 only because rib presence is itself labeled.

194 Sixteen records carry a lumbar rib in the released volumes: thirteen bilateral and three
 195 unilateral, all three on the right. That laterality is reported because it is what the release
 196 contains and not as a finding — sixteen records cannot support a statement about side
 197 preference, and none is made.

198 Eighteen records carry an L6. Fourteen are labeled lumbarization, two sacralization,
 199 one semi-sacralization and one normal, which is worth stating because it shows the two
 200 annotation axes disagreeing in the open: a record can carry six lumbar-type vertebrae and
 201 still be read as a sacralization, or as unremarkable, depending on where the reader started
 202 the count.

203 Both counts are derived from the released label volumes, and so are the manifest fields
 204 that report them. That is a correction rather than a design note: through v5 the `has_16`
 205 flag was true for exactly one record, which contains no L6, and false for all eighteen that
 206 do, while `has_lumbar_rib` was false everywhere and `n_lumbar_labels` read zero in 799 of

207 802 records. A reader selecting the six-lumbar cases on those fields would have received one
208 record that is not one, and a reader selecting lumbar ribs an empty set, with nothing to
209 indicate either. The fields are rebuilt for this release by counting identifiers 25 and 74–75 in
210 each volume, and `lumbar_rib_side` is added alongside them.

211 II.F. Computational tools

212 *Access.* Every processing step described above is performed by code released under an
213 open-source licence and archived with the dataset; no step depends on software that is
214 not publicly obtainable. The release archive carries the reference loader, the label-scheme
215 definition and the quality-control scripts that generated the tables in this manuscript.

216 *Versions and parameters.* Segmentation of femora, the sacral sub-division and the per-
217 level rib numbering used TotalSegmentator¹⁶ (`total` task, release 2.x) on a single graphics
218 processing unit. The binary rib network is Möller’s¹⁵ released weights, applied through the
219 nnU-Net v2 framework¹⁸. The pelvic completion for records whose pelvis was absent used
220 a five-fold nnU-Net v2 ensemble, applied out-of-fold: each record is completed by the fold
221 that never trained on it, so no record is completed by a model that has seen it. The five
222 fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the
223 data (<https://huggingface.co/anonymous-mlhc/spinopelvic-seg-checkpoints>). Im-
224 age handling throughout used NiBabel and SciPy; no image is resampled or reoriented when
225 a label is written, so every label shares its image’s grid and affine exactly.

226 *Generative artificial intelligence.* Large-language-model assistance was used for manuscript
227 preparation — drafting and editing of text, and generation of analysis and figure code —
228 under author review. It was not used to generate, annotate or interpret any imaging data,
229 and no scientific claim in this manuscript originates from it. All numbers reported here are
230 computed by the released code from the released volumes.

231 II.G. Validation

232 Validation is layered, and the ordering matters: measurements computed on a corpus
233 that has not established internal consistency do not look broken, they look finished.

234 *Geometric invariants.* Every case is checked for label geometry agreeing with its CT in

shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. The check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed a label away from its image, a failure invisible downstream.

The check compares *each sided pair separately*, and both simpler alternatives miss cases it catches. Testing the ribs alone passed all 802 records while four of them carried `left_hip` on the patient’s right. Pooling all sided structures into one test recovered three of those four and missed the fourth, because hips and femora outweigh the ribs by voxel count, so a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib–vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance; of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is misnumbered. The denominator is stated because quoting the two residual offsets against all 11,548 ribs would halve the rate by counting ribs the check never examined — and because more than half the ribs in a screening abdominal CT being unevaluable is itself a fact about the corpus worth reporting. Both offsets are field-of-view truncations on individually characterized cases, one at +1 level and one at −1, and are reported rather than suppressed: a threshold tuned until the count reaches zero yields a cleaner number and a less informative one.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra — not because no lumbar ribs exist, since 16 records carry one, but because such a rib takes class 74 or 75 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

Agreement with published values. Derived spinopelvic measures are compared against Vialle *et al.*¹⁹ (Table II). Pelvic incidence is the strongest of these checks because it is a morphological property of the pelvis rather than a posture, and so is directly comparable to a standing reference cohort.

TABLE II Derived measures against a standing asymptomatic reference ($n = 260$)¹⁹. This cohort is supine: sacral slope is 4.1° lower and pelvic tilt 4.3° higher, one discrepancy rather than two, since $PI = PT + SS$.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI – LL mismatch	2.6°	centered near 0

II.H. Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade²⁰ in the released manifest: 33 cases, typed I–IV with the a/b unilateral–bilateral qualifier. This is released as an annotation layer because the grade and the count are *different axes* and the dataset is built to show that they are: grade IIIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a perfectly normal count of five. A corpus that recorded only the count would describe those seven as unremarkable.

Two radiology resident physicians graded every case independently and resolved their seven disagreements by consensus; both reads and the consensus are in the manifest, with one normal reference read. The distinction the classification turns on — bony fusion against articulation without it² — is the one carrying clinical weight, and it is where the disagreements lay: four of the seven were III against II.

II.I. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**: a cage-bridged interspace reads as “no gap” exactly as a fused transitional vertebra does. An unrecorded instrumented case is therefore a false positive on the central measure.

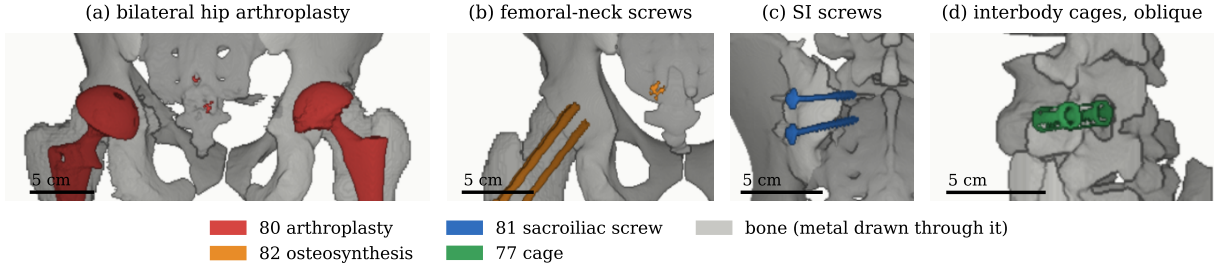


FIG. 3 The four instrumentation types in the release, one exemplar each, rendered from the released labels. (a) Hip arthroplasty, identifier 80, $n = 8$ records (this one bilateral). (b) Femoral-neck osteosynthesis, 82, $n = 1$. (c) Sacroiliac screws, 81, $n = 1$. (d) Interbody cages at L5–L6, 77, $n = 1$, oblique view. The implant is drawn through the bone that would otherwise hide it. Each panel is framed on its implant; the bar is 5 cm.

284 *II.I.0.a. Detection over-calls by a factor of five.* A 2500 HU threshold inside a shell
 285 around the labeled skeleton flags 52 of the 802 records. One of the two radiology resident
 286 physicians read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and
 287 reconstruction artefact. Attenuation does not separate the two groups (both saturate);
 288 site and volume do, every confirmed implant at $2,586 \text{ mm}^3$ or above and every rejection at
 289 $1,768 \text{ mm}^3$ or below. Instrumentation prevalence is 1.4%, not the 6.5% the threshold alone
 290 reports.

291 *II.I.0.b. The subtype block had to be extended.* Identifiers 76–79 name spinal instrumen-
 292 tation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so
 293 **80 arthroplasty**, **81 sacroiliac screw** and **82 osteosynthesis** were added: osteosynthe-
 294 sis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule
 295 calls a femoral stem a rod. The confirmed cases are 8 arthroplasties, one osteosynthesis, one
 296 sacroiliac fixation and one pair of interbody cages (Fig. 3).

297 *II.I.0.c. Metal outranks bone, and that is a subtraction.* In every confirmed case the
 298 implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than
 299 adding them. Pelvic incidence and tilt are measured from the femoral head center, and in
 300 8 of these cases that head is an implant: any spinopelvic parameter previously computed
 301 from them was measured on metal.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canonicalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM: the policy prefers DICOM where applicable, and for volumetric segmentation masks NIfTI is what the downstream tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the data as `splits_5fold.json`. The five validation folds partition all 802 records, so the release provides cross-validation and *not* a held-out test set. Users reporting a single held-out number should carve one out and state which records it holds, because the folds shipped here will not do that job.

Stratification is on the transitional subtype rather than on a binary LSTV flag, and it enforces a minimum of each rare subtype in every validation fold: each fold’s validation set carries three sacralization and three lumbarization records, so no fold is evaluated on a cohort missing the class the dataset exists for. With 15 lumbarization and 15 sacralization records across 802, that is the practical limit of what stratification can guarantee, and it is why a fold-level metric on the rare classes carries an error bar far wider than its own decimal places.

Table III lists the key data types and metadata against the fraction of records for which each is populated.

The archive of record is Zenodo (10.5281/zenodo.22139642). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value, a recorded value and not a missing one, and 53 carry no sex field. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals, and records outside the female/male dichotomy are excluded from sex-stratified measures — a limitation of those measures rather than of the cohort, stated because folding eleven people into “unrecorded” would misdescribe the source. The lower

TABLE III Key data types and metadata, and the fraction of the 802 records populating each, counted from the released manifest. Below 100% is not an error: age and sex are absent where the source did not publish them, a pelvic series identifier exists only where a pelvic annotation was placed, and a Castellvi grade only for graded records.

Field	Records	Available
Identifier, image path, label path, configuration, match type	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class	802	100.0%
Spine and pelvic annotation origin	802	100.0%
Instrumentation and partial-annotation flags	802	100.0%
Image/label alignment check	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade (consensus of two readers)	34	4.2%

age bound is the screening guideline rather than a selection applied here, and it is why the measures below sit closer to a reference range than a surgical series would.

Count-free measures (Fig. 4). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal.

Level gradients and spinopelvic measures (Fig. 5). Vertebral dimensions increase caudally in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic tilt increases — the descriptive signature of a pelvis retroverting as a spine ages.

Opportunistic measures. Every scan carries a vertebral bone density measurement at no additional dose.²¹ L1 trabecular attenuation is reported at the published standard site.

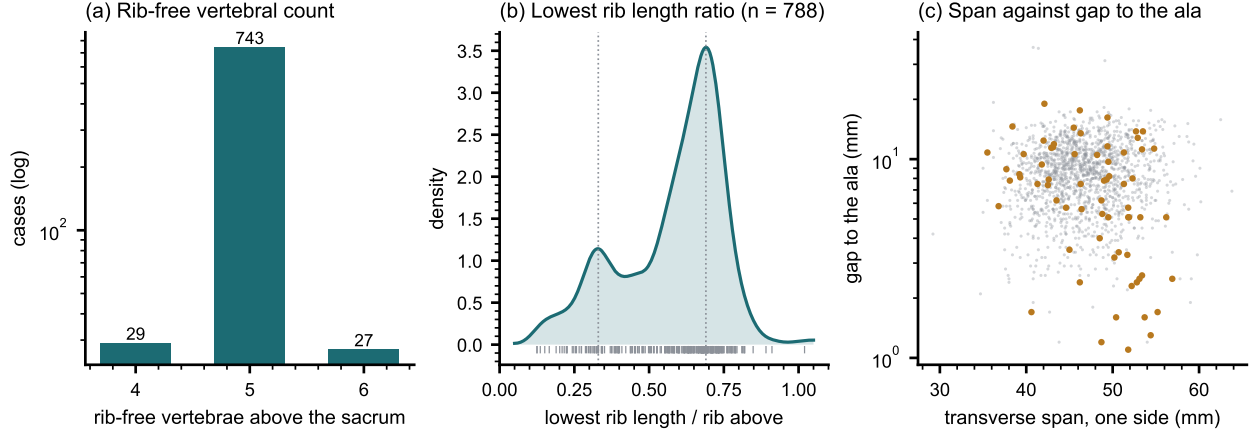


FIG. 4 Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b) Lowest rib length as a fraction of the rib above: two populations, near 0.33 and 0.69. (c) Lowest lumbar transverse span against its gap to the sacral ala; source transitional labels marked.

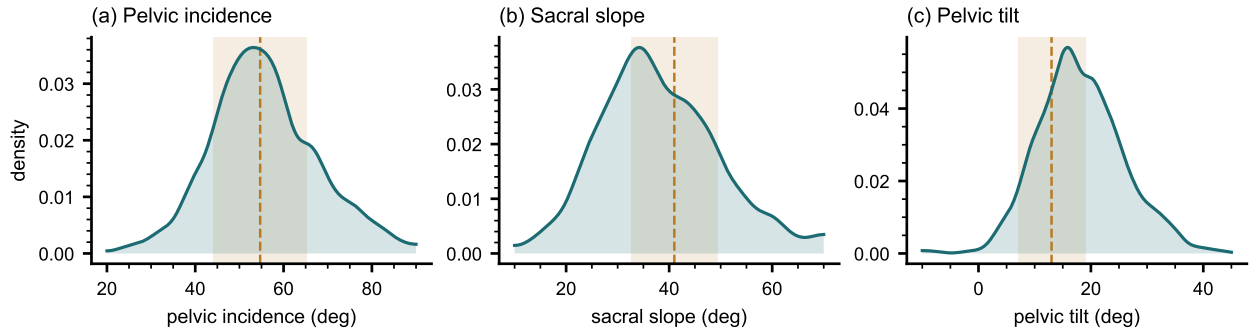


FIG. 5 Spinopelvic measures against standing reference bands (mean $\pm$ SD)¹⁹. Sacral slope is postural and this cohort is supine; pelvic incidence is not, and its agreement is the strongest check.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*: spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, which makes sacral morphology measurable against the lumbar column rather than in isolation — the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads rather than from landmarks placed on a radiograph, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it: no trajectory can be planned across a junction whose segments cannot be named. Level-numbering benchmarks,

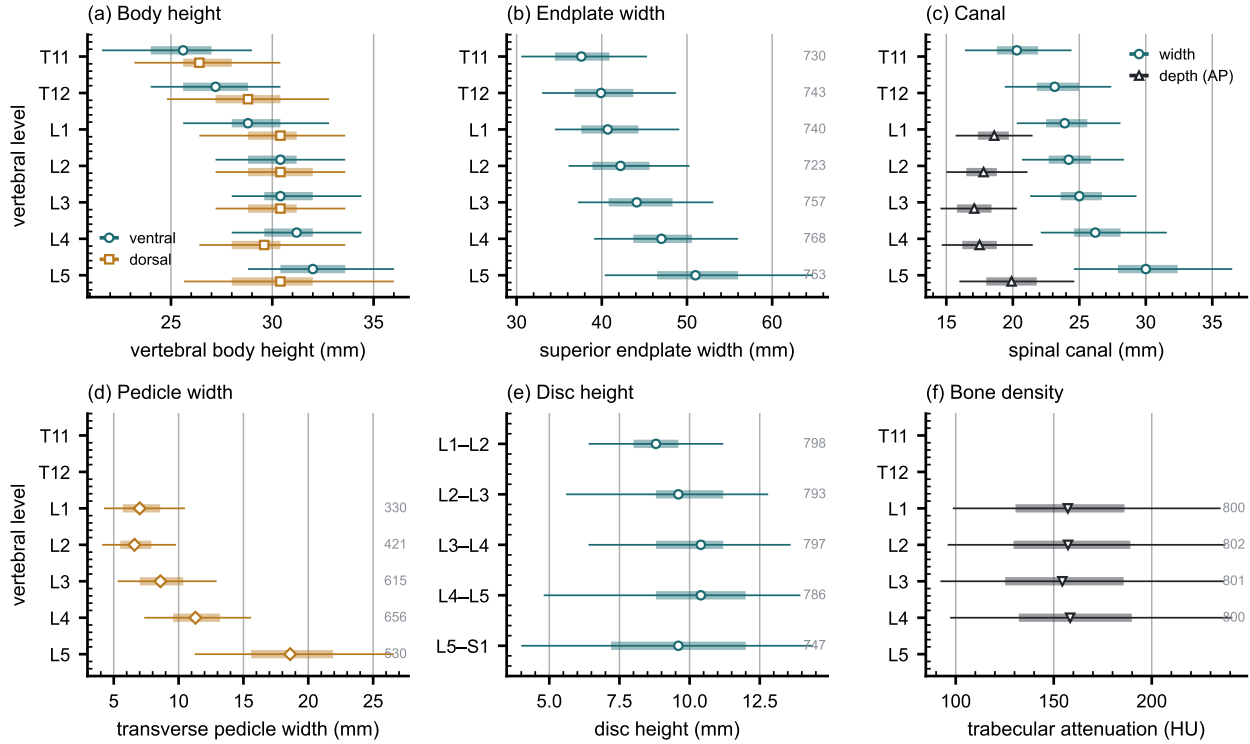


FIG. 6 Lumbar and lower thoracic morphometry by level. Marker, median; bar, interquartile range; line, 5th–95th percentile; n at right. (a) Body height, ventral and dorsal. (b) Superior endplate width. (c) Canal width and depth. (d) Pedicle width. (e) Disc height at its interspace. (f) Trabecular attenuation at the opportunistic-screening site. Canal depth and pedicle width rest on 330–656 records, the others on 720–784. Intervals are dispersion; no comparison between levels is implied.

variant studies and opportunistic screening follow from the same annotation.

V.A. Three uses the released measurements support

Naming a level from its own geometry. Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this cohort was assembled around.²² The failure is structural rather than careless: the count is ambiguous exactly where the variant sits. Per-level body height, canal width, endplate width, transverse-process span and wedge ratio are released for T11–L5 across all 802 records, and because the labels were assigned against the twelfth rib rather than by enumeration, a model trained on them is not learning the convention that fails.

360 *Reference morphometry with its spread.* The values a surgeon consults for vertebral
361 body height, canal dimensions and pedicle width come from cadaveric series of a few dozen
362 specimens, plotted as single values with no indication of spread.¹⁰ They cannot tell a reader
363 whether the patient in front of them is unusual, because they never show what usual looks
364 like. The same measures are given here from 802 records with the distribution attached
365 (Fig. 6), reproducing two properties of the classical figures — body height crosses over,
366 dorsal exceeding ventral at T11–T12 and reversing by L4–L5, and pedicle width widens
367 steeply into the lower lumbar spine — and adding the width of each distribution, which is
368 the part needed to judge an individual. A larger n sharpens a distribution without changing
369 the population it came from, and this one is a screening cohort aged 50 and over.

370 *Patient-specific models.* Planning is moving toward patient-specific biomechanical models
371 rather than population norms.²³ Such a model needs spine, pelvis and femoral heads in
372 one frame, which is this release’s construction; in 351 patients the annotation sits on two
373 acquisitions in different positions, giving a within-patient change in alignment against which
374 a predicted postural response can be checked.

375 Its limitations are part of it, and they are stated at the level of detail a reader would
376 need to catch them independently.

377 *Coverage.* Thoracic ground truth is field-of-view limited (Fig. 7) and does not extend to
378 T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat.
379 The cohort is a colorectal screening population aged 50 and over, so its distributions should
380 not be read as representative of a surgical one.

381 *Declared but empty, and confirmed so.* The partial-annotation sentinel (255) is declared
382 in the scheme and occurs in no record — established by counting identifiers across all 802
383 released volumes rather than asserted. It is retained because the protocol is part of the
384 scheme and a later release may ship a partially traced record. The hardware identifiers were
385 in this position until v6 and are no longer: 76–82 are populated in the eleven instrumented
386 records described in Sec. II.I.

387 *Strength of the labels varies by structure, and the release does not average over that.*
388 Vertebral labels derive from radiologist-supervised source annotations, corrected where those
389 sources were wrong. Pelvic labels on records that lacked one are pseudolabeled. The rib
390 layer is a pseudolabel whose human review was triaged by an automated rule rather than
391 exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional

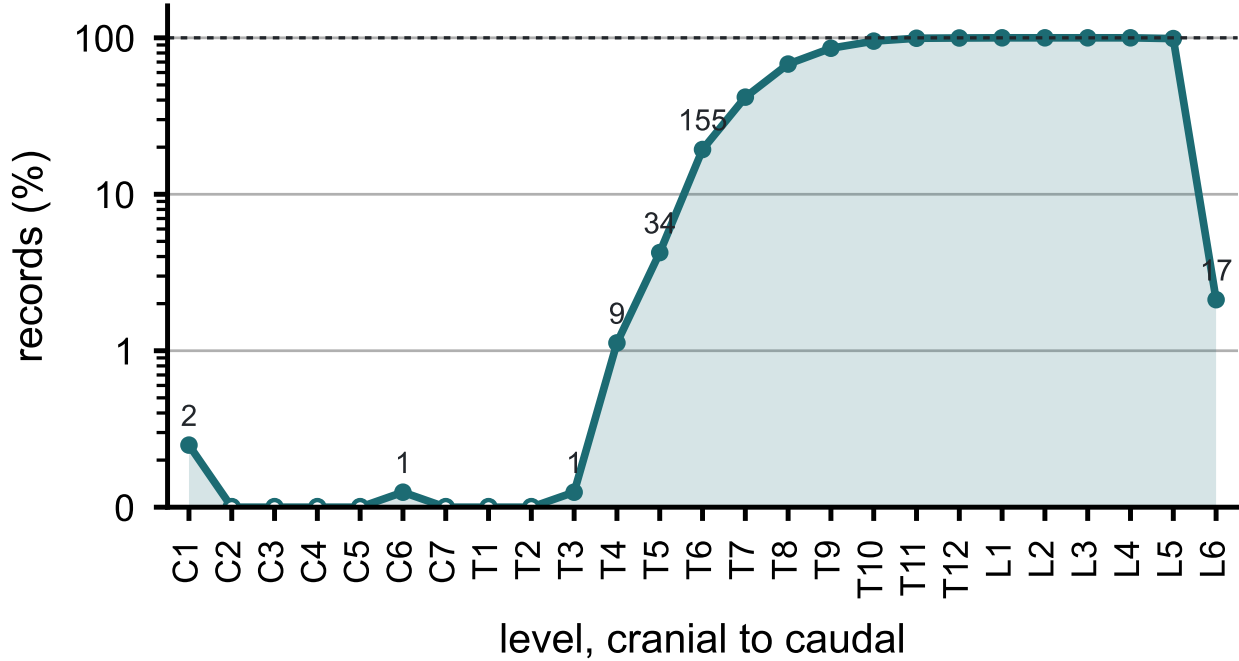


FIG. 7 Records carrying each vertebral level, of 802, cranial to caudal: the field of view of the cohort as the labels see it. The thoracic spine enters the abdominal field of view from below. Per-identifier counts are in Table S1.

labels derive from two independent sources and are not uniformly adjudicated — which is precisely why the measures reported here are count-free — and the Castellvi grades are a consensus of two radiology resident physicians. The sacral sub-division inherits its level from an automatic method, and in 100 records that level places S1 above half the sacrum or below fifteen percent of it, which no first sacral segment is; those records are flagged in the released quality-control table and should be filtered before any measurement depending on the sacral endplate.

Structures are not universally present. The same census shows the release is not uniformly complete. Sacrum, hips and femora are present in all 802 records. One record carries no S1, one lacks T12, and nine lack an L5 identifier — in every case because the structure lies outside the field of view or, for L5, because the lowest lumbar segment is labeled L6 or incorporated into the sacrum. Users filtering on the presence of the caudal anchor should filter explicitly rather than assume it.

405 VI. DISCUSSION

406 *Comparison with related material.* Against the collections in Table I, the contribution
407 is the crosswalk placing spine and pelvis on the same series, together with classes for the
408 anatomy an enumeration anomaly actually produces.

409 VII. CONCLUSION

410 CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame
411 in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of
412 its own rather than recording it as something it is not. Because both counting anchors are
413 explicit, a level can be identified from a field of view that cannot support the conventional
414 count downward from C2 — which is every record here.

415 DATA AVAILABILITY

416 The release described here is v7, permanently archived at [DOI redacted for review]; the
417 concept DOI [DOI redacted for review] resolves to the current version. The archive carries
418 the loader, the label scheme, the quality-control tables and the analysis code under an open-
419 source licence. The repository URL identifies the authors; it is on the title page and will be
420 restored here for the camera-ready version.

421 CONFLICT OF INTEREST

422 The authors have no relevant conflicts of interest to disclose.

423 ETHICS

424 This work uses publicly available, de-identified imaging and annotations derived from it.
425 The authors’ institutional review board determined that it is not human participant research
426 under 45 CFR 46 and requires no oversight.

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